

BUYERS' GUIDE

ESTYLE

The Best Wi-Fi ROUTER For Your Office



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If you run a small business, it is advisable to hand-pick your own Wi-Fi router. While the router provided by your Internet service provider may be authentic, it might not be the right fit for your business. Routers across the globe have come a long way with significant improvements in speed capabilities, security and signal strength consistency.

Router speed, signal strength and bandwidth are the key priorities for home offices. However, small businesses require a few more features as an integral part of the setup. You need to find the best match for your needs.

WPA2 AES security

Not familiar with the jargons WPA or AES? Let us explain. Wi-Fi Protected Access (WPA) is a wireless network security protocol. For small offices or home office setups, WPA2 utilising 128-bit Advanced Encryption Standard (AES) with pre-shared key (PSK) encryption compatibility is the most popular as well as preferable algorithm with disabled Wi-Fi Protected Setup (WPS). Bigger businesses go for WPA2-Enterprise, also known as RADIUS security protocol, which provides the best level of security. But the associated cost, infrastructure and maintenance requirements might not make it feasible for a small business arrangement. Other than this, good practices and security measures like active internal firewall and updated firmware can enhance the network safeguard.

Extra ports

Home routers usually come with four 10x100Mbps LAN ports and one WAN port. Small businesses should look for routers with extra ports. Else, they can pick a switch to expand their networks. To expand bandwidth or ensure a working port as a backup for connectivity if the main WAN malfunctions, look for extra WAN ports to connect additional service from your ISP. That way, you can balance the load on the network when your business supports a bigger employee base.



What you need

Security standard: WPA2 AES PSK
Scalability: Preferably additional LAN and WAN ports apart from the general four and one
Network: Dual-band
Wi-Fi standard: 802.11ac or at least 802.11n
Antenna: At least 2x2
IP version: IPv6

Dual band to avoid network jamming

Networks get painfully slow when buffering a video or opening a heavy website. That happens due to cluttering of network channels. Single-band Wi-Fi routers function at a bandwidth frequency of 2.4GHz, which provides a good coverage range; the higher the frequency value, the lower the range. But this specific bandwidth is also tapped in by other gadgets like phones and tablets. Lower channel width is jammed by multiple devices, which causes the 2.4GHz bandwidth to clutter faster. Therefore high-volume data streaming becomes slow for single-band routers.

Dual-band routers can tap into two frequency bandwidths: 2.4GHz and 5GHz. While 5GHz bandwidth provides much smaller coverage area, the wider data channel allows it to transmit much larger volumes of data. Latest dual-band routers divide their data flow—maintaining the lower-volume data through the 2.4GHz frequency, while routing the higher data volume, for example in video stream or gaming, through the 5GHz frequency.

Latest 802.11 standard for best performance

The IEEE Standards Association has introduced Wi-Fi standards for high network throughput. The most efficient standard at the moment is 802.11ac, which supports data throughput (data transmission rate) up to 2Gbps (depending on the number of spatial streams, each of which carries around 433Mbps of data), wireless range up to 61 metres (200 feet), and four wireless channels of